Vijiya Gupta

New York, NY

(705) 976 - 6157 | vijiya.gupta@suny.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering in Financial Engineering | **GPA: 3.83/4.0** **Expected Dec. 2025**

Relevant Coursework: Asset Management, Behavioral Finance, Derivatives, Fixed Income, Machine Learning, Market Microstructure, Portfolio Management & Quant Strategies, Stochastic Calculus, Time Series Analysis, Volatility Trading
Leadership Experience: Team lead for the CFEM Capstone Project with Marathon Asset Management

Indian Institute of Technology (IIT) Goa, Goa, India
Bachelor of Technology in Mathematics and Computing | **GPA: 8.45/10** **May 2024**
Leadership Experience: Branch student mentor; Design lead of the Institute Newsletter

SKILLS

Python (Pandas, NumPy, Scikit-learn, Matplotlib, PyTorch, spaCy), SQL, R, MATLAB, Power BI, Excel, LaTeX, NetLogo
Data Visualization, Statistics, Analytics, Predictive Models, Financial Markets, Natural Language Processing (NLP), Machine Learning (ML), Time-Series Forecasting, Econometrics, Language Models

EXPERIENCE

Cornell University, *Graduate Research Assistant,* Ithaca, NY **Jan. 2025 - Present**

- Supported PhD research on firm-level **financial data** (Factiva, WRDS), georeferencing FEMA maps (QGIS), and analyzing macro-climate indicators and sustainability disclosures.

Alliance Bernstein*, Data Science Intern,* Remote **Jun. – Aug. 2025**

- Engineered an unsupervised **LLM** framework for interpretable factor extraction from news using **LLaMA** models.
- Implemented rolling-window context, autoregressive updates, and reflection-based reasoning for incremental performance gains.
- **Backtested** across sectors, achieving above-random accuracy in next-day excess return prediction.
- Performed event studies and diagnostic analyses to evaluate **portfolio impact, model reasoning behavior, and sector-specific feature relevance.**

Genpact*, Data Science Intern – Media Analytics,* Bangalore, India **Aug. - Nov. 2023**

- Measured advertisement campaign efficacy amidst third-party cookie deprecation **via Ridge Regression, lift measurement, and propensity scoring**, thereby improving targeting accuracy.
- Designed a **bias-correction method** combining DenseWeight and Jaccard's distance to address sparse dataset representation, reducing error margins by 10%.

Adobe*, Product (ML) Intern - Core Tech & Cloud Ops,* Bangalore, India **May – Jul. 2023**

- Developed an **anomaly detection prototype** leveraging Prophet to forecast operational metrics and reduce false positives in alerting systems.
- Built and deployed a **Kedro-based** data pipeline on **AWS** Lambda to automate early fault detection, reducing developer troubleshooting time.

PROJECTS

Building a Private Credit Index for Alternative Lending Markets, *CFEM x Marathon Asset Management* **Ongoing**

- Building an open-source **private credit index** using publicly available SEC 10-Q/10-K filings from Business Development Companies (BDCs), representing **300K+ individual loan-level investments**.
- Automated XBRL scraping and parsing of SEC filings; performed exploratory analysis and extensive data cleansing on sparse and inconsistent disclosures.
- Designing a hybrid **rule-based and LLM-driven data engineering workflow** to reconcile missing/incorrect loan attributes, improving data usability and reducing manual intervention.
- Constructing an **index** and **sub-indices by industry, debt instrument, and loan characteristics**, with reusable and modular code for future scaling.

ESG-Aware Portfolio Optimization with Regularization and Clustering, *Cornell University* **Feb. - May 2025**

- Developed a Sharpe-maximizing portfolio framework integrating ESG and emissions constraints using S&P 500 data.
- Applied clustering (ESG, Scope 3, Sharpe), L2 regularization, and thresholds to improve diversification. Evaluated tradeoffs between sustainability and out-of-sample performance.